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To the Editor-in-Chief and Associate Editor
IEEE Transactions on Information Theory

Dear Editor:

Please consider the enclosed manuscript, “Tradeoffs Between Rate and Conditional Content with Encoder-Observed Context,” for publication in the IEEE Transactions on Information Theory. I am the sole author. The review manuscript is in single-column format.

The paper is the first exactly solved instance of a broader question, how the information cost of a description depends on which party observes what, and it is written to stand on its own in classical rate–distortion terms.

The paper evaluates Steinberg’s common-reconstruction functional on an augmented Gaussian source: the encoder observes a pair (Y, V) , the distortion binds Y alone, and a third party holds a noisy copy of the context V that no decoder uses. Three results are primary: a closed form for the minimal conditional content, the larger root of an explicit quadratic, for Y and V scalar linear functionals of an arbitrary-dimensional jointly Gaussian source; the exact region of jointly achievable rate and conditional content, with a two-noise-fraction Pareto frontier and a strict tradeoff exactly when the described variable and the context are partially correlated, persisting at the clean-context boundary. A structural consequence is non-determination: the function is not a functional of the reduced source–side-information marginal, so the encoder’s access to the context changes the operational problem. The third primary result extends the closed form to a described vector with independent coordinates (Theorem 25): the content-optimal split of the distortion is not reverse water-filling, and the rate-optimal and content-optimal descriptions can describe disjoint sets of coordinates. For coupled coordinates the function is a determinant maximization in the error covariance left to the side-information holder (Theorem 28), explicit in a one-dimensional regime as a generalized eigenvalue whose scalar case is the closed form (Theorem 29), and there the direction the content-optimal description reads turns with the distortion budget unless it ignores the context (Theorem 30). Extensions give the frontier with vector context, the binary rate–content frontier through a tilt equation, and a direct Gaussian operational theorem whose achievability quantizes the conditioner, the source, and the reproduction and invokes the finite-alphabet theorem. Consistency anchors recover the classical, Gray, and Steinberg corners.

Two earlier documents are relevant, and they are distinct from one another. First, this Transactions declined a different manuscript of mine, IT-26-0996, on 27 July 2026 without review, on scope and presentation; it was a broad synthesis. The present submission is materially distinct: it is a focused source-coding paper, shares no theorem with IT-26-0996, and uses standard source-coding terminology throughout.

Second, and separately, the manuscript cites an archived preliminary manuscript by the author, reference [36] of the submission (Zenodo record, doi:10.5281/zenodo.21776291), as disclosed in its Related Work section. That record is not a prior publication of this paper, and the two are distinguished as follows. Inherited from [36], with attribution: the discrete operational region theorem and its single-letterization, and the scalar Gaussian corner. New in the present manuscript, and not in [36]: the closed form $L(D) = \frac{1}{2} \log_2 g^*$ with its achieving channel for Y and V scalar linear functionals of an arbitrary-dimensional Gaussian source (Theorem 14(a)), the exact rate–content region with its two-noise-fraction Pareto frontier (Theorem 14(b)), the misalignment and non-determination corollaries (Corollaries 19 and 21), the vector-context extension, the parallel and coupled described-variable results (Theorems 25 and 28–30), the binary rate–content frontier, the direct Gaussian operational theorem (Appendix A), and the Lu–Xu comparison (Appendix C). Two components have close predecessors that are credited in place: the pair-sufficiency reduction is a distortion-preserving specialization of Armstrong’s information-bottleneck sufficiency, and the binary conditional-content function coincides with the common-reconstruction rate of Lu et al., the contribution there being the joint frontier. The present paper supersedes the source-coding content of [36], which remains an archived evidence artifact. The present manuscript’s own sources, verification scripts, and reproduction notebook are archived separately (doi:10.5281/zenodo.22678341) and at a tagged path of the public repository named in the title footnote.

The closed forms have been checked numerically against direct minimization; the check scripts and an executed reproduction notebook are submitted as supplementary material. The use of AI tools in preparing the manuscript is disclosed in its Acknowledgment, as IEEE policy requires.

This manuscript is not under review elsewhere, and I have no conflicts of interest to declare.

Sincerely,

Andrew H. Bond
Senior Member, IEEE